Homework, Lecture 5: Tacit Collusion in Oligopoly

Solution — Economics of Organizations and Incentives (MPE Insper)

November 2026

The problem

Source. Lecture 5 slides, `Aula/Lecture05.pdf`, slide 34 (Homework: tacit collusion); no answer given. Reproduced verbatim.

> Two firms play the Cournot game from Lecture 2 repeatedly, forever, with discount factor δ: inverse demand $P = 100 - q_1 - q_2$ and cost $C(q_i) = 10\,q_i$.
>
> 1. Find the Cournot-Nash outputs and profits of the stage game. Find the cartel outcome, in which each firm produces half of the monopoly output, and the profit of each firm.
> 2. Suppose the rival produces the cartel output. What is the best deviation, and the profit it yields?
> 3. Write the grim-trigger strategy: produce the cartel output as long as both firms have done so; otherwise, produce the Cournot-Nash output forever. Use the one-stage deviation principle to find the values of δ for which it is a subgame-perfect equilibrium.
> 4. Show that the threshold does not depend on the demand and cost parameters: redo item 3 with $P = a - b(q_1 + q_2)$ and marginal cost $c < a$.
> 5. Which market features make tacit collusion easier? Relate them to item 3.

Notation (Lecture 5). A history h is the list of past action profiles; $h = \varnothing$ at the start. A strategy s_i maps each history into a quantity. Payoffs are normalised, $U_i(s) = (1-\delta)\sum_{t\geq 1}\delta^{t-1}u_i(a^t(s))$ (slide 21), so a constant stream k is worth k. $s|_h$ is the continuation strategy at h. The one-stage deviation principle (slide 25): s is an SPE iff for every history h, player i and action a_i,

$$U_i(s|_h) \;\geq\; (1-\delta)\,u_i\big(a_i, s_{-i}(h)\big) + \delta\,U_i\big(s|_{(h,(a_i,s_{-i}(h)))}\big).$$

The method is that of grim trigger in the prisoner's dilemma (slides 28–29), where it gives $\delta \geq 1/4$: check the cooperative state, then the punishment state.

1 Item 1: stage-game Nash and cartel outcomes

Firm i solves $\max_{q_i}(100 - q_i - q_j - 10)\,q_i$. The first-order condition $90 - 2q_i - q_j = 0$ gives the best response

$$BR_i(q_j) = \frac{90 - q_j}{2}.$$

In the symmetric Nash equilibrium $q = (90-q)/2$, so $q_N = 30$. Industry output is 60, the price is 40, and each firm earns

$$\pi_N = (40 - 10)\cdot 30 = 900.$$

Cartel. A monopolist maximises $(100 - Q - 10)Q$: $90 - 2Q = 0$, so $Q^m = 45$, the price is 55 and the monopoly profit is $(55 - 10) \cdot 45 = 2025$. Each firm produces half, $q_C = 22.5$, and earns

$$\pi_C = \frac{2025}{2} = 1012.5 \; > \; \pi_N = 900.$$

Answer. Cournot-Nash: $q_N = 30$ each, price 40, $\pi_N = 900$. Cartel: $q_C = 22.5$ each (total 45), price 55, $\pi_C = 1012.5$.

2 Item 2: the best deviation

If the rival produces $q_C = 22.5$, firm i maximises its stage profit against it:

$$q_D = BR(22.5) = \frac{90 - 22.5}{2} = 33.75.$$

Industry output is $33.75 + 22.5 = 56.25$, the price is 43.75 and

$$\pi_D = (43.75 - 10) \cdot 33.75 = 33.75^2 = 1139.0625.$$

The ranking $\pi_D > \pi_C > \pi_N$ $(1139.0625 > 1012.5 > 900)$ is the prisoner's-dilemma structure of the stage game: cartel output is not a stage-game Nash equilibrium, and each firm gains $\pi_D - \pi_C = 126.5625$ by cheating once. Being punished costs $\pi_C - \pi_N = 112.5$ per period.

Answer. Best deviation $q_D = 33.75$, profit $\pi_D = 1139.0625$.

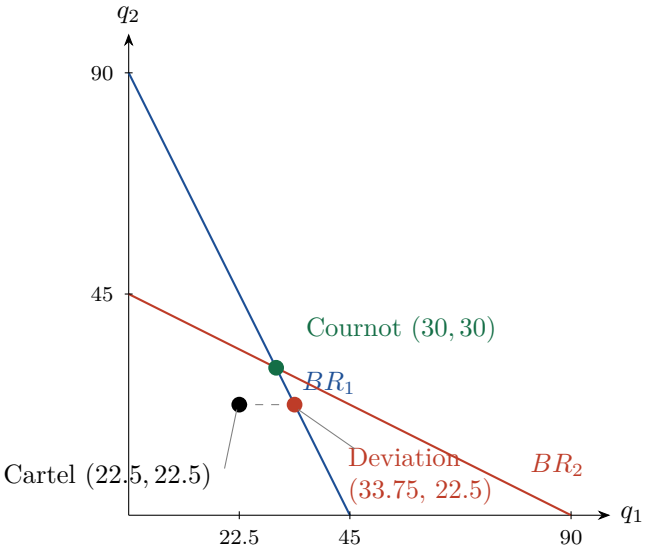


Figure 1: Best-response functions. The deviation point lies on firm 1's best response to the cartel output $q_2 = 22.5$. Payoffs (π_1, π_2): cartel $(1012.5,\ 1012.5)$, deviation $(1139.0625,\ 759.375)$, Cournot $(900,\ 900)$.

3 Item 3: grim trigger and the one-stage deviation principle

Strategy. Let h_C^t denote a history in which both firms produced q_C in every past period (and $h = \varnothing$). For firm i,

$$s_i(h) = \begin{cases} 22.5 & \text{if } h = \varnothing \text{ or } h = h_C^t, \\ 30 & \text{otherwise.} \end{cases}$$

There are two states, as in slide 28: *collusion* (both have always produced q_C) and *punishment* (anyone ever deviated; absorbing, so cartel output never resumes).

Collusion state. Following s gives π_C forever, so $U_i(s|_h) = \pi_C$ (normalised). The best one-stage deviation is q_D, which yields π_D today and moves play to the punishment state, worth π_N per period. The constraint is

$$\pi_C \;\geq\; (1-\delta)\,\pi_D + \delta\,\pi_N.$$

Equivalently, in unnormalised form, $\dfrac{\pi_C}{1-\delta} \geq \pi_D + \dfrac{\delta\,\pi_N}{1-\delta}$. Rearranging, $(1-\delta)(\pi_D - \pi_C) \leq \delta\,(\pi_C - \pi_N)$, so

$$\delta \;\geq\; \delta^* = \frac{\pi_D - \pi_C}{\pi_D - \pi_N} = \frac{126.5625}{239.0625} = \frac{9}{17} \approx 0.529.$$

(Here $\pi_D - \pi_N = 1139.0625 - 900 = 239.0625$.) The numerator is the one-period gain from cheating; the denominator is that gain plus the per-period loss from punishment. So δ^* is the share of the total at stake that the future must outweigh. At $\delta = 1/2$: $\pi_C = 1012.5$ against $\frac{1}{2}\cdot 1139.0625 + \frac{1}{2}\cdot 900 = 1019.53125$, so cheating pays. At $\delta = 0.6$ cooperation wins.

Punishment state (off the path). After any history other than h_C^t, s prescribes $q_N = 30$ forever; the continuation payoff is $\pi_N = 900$. A one-stage deviation to x against a rival who plays 30 gives $(1-\delta)(100 - x - 30 - 10)x + \delta \cdot 900$, and the stage profit $(60 - x)x$ is maximised at $x = 30$, where it equals 900. So no one-shot deviation is profitable, for any δ: the Cournot-Nash profile is a Nash equilibrium of the stage game, and playing it after any history is credible. This is the argument of slide 29 ("which always holds: (F, f) is a Nash equilibrium of the stage game").

Conclusion. The only binding constraint is the collusion state. By the one-stage deviation principle, grim trigger is a subgame-perfect equilibrium iff $\delta \geq 9/17$. For $\delta < 9/17$ the cartel output is not sustainable by this punishment.

Answer. Grim trigger is an SPE iff $\delta \geq \delta^* = 9/17 \approx 0.529$.

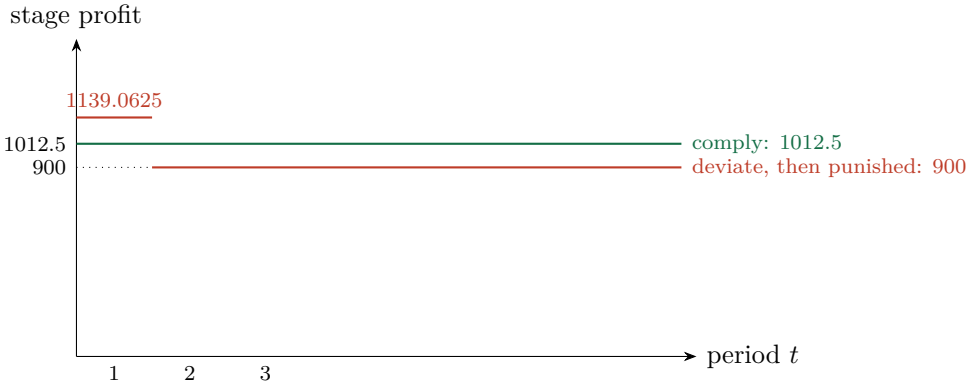


Figure 2: Stage profits under compliance and after a deviation in period 1 (vertical axis not to scale from zero). The deviator gains 126.5625 once and loses 112.5 in every later period; discounting decides which dominates, and the break-even is $\delta^* = 9/17$.

4 Item 4: general demand and cost

Let $P = a - bQ$, $Q = q_1 + q_2$, with $b > 0$ and constant marginal cost c, $0 \leq c < a$. The plan is to redo items 1–3 symbol by symbol and watch a, b and c collapse into one scale factor,

$$k \;=\; \frac{(a-c)^2}{b} \;>\; 0.$$

k is positive because $c < a$ and $b > 0$. Positivity matters in Step 6, where we divide an inequality by k.

Step 1: stage profit and the best response

Firm i's stage profit when the rival produces q_j is

$$\pi_i(q_i, q_j) = (a - b(q_i + q_j))q_i - c\,q_i = (a - c - bq_j)q_i - bq_i^2.$$

It is a concave parabola in q_i: $\partial^2\pi_i/\partial q_i^2 = -2b < 0$, so the first-order condition gives the maximum:

$$\frac{\partial \pi_i}{\partial q_i} = a - c - bq_j - 2b\,q_i = 0 \quad \Longrightarrow \quad BR(q_j) = \frac{a - c - b\,q_j}{2b} \qquad (\text{for } q_j \le (a-c)/b).$$

A shortcut used below. Rewrite the first-order condition as $a - c - b(q_i + q_j) = b\,q_i$. The left side is the price-cost margin $P - c$. So, *at any best response*, the margin equals bq_i, and profit is margin times output:

$$\pi_i(BR(q_j), q_j) = (P - c)\,q_i = b\,q_i^2.$$

This holds for the Cournot output and for the deviation, both of which are best responses, but *not* for the cartel output, which is not one.

Step 2: the Cournot-Nash equilibrium

In a symmetric equilibrium $q_1 = q_2 = q$ with $q = BR(q)$:

$$2b\,q = a - c - b\,q \implies 3b\,q = a - c \implies q_N = \frac{a-c}{3b}.$$

Industry output is $Q_N = \dfrac{2(a-c)}{3b}$, so the price and margin are

$$P_N = a - b \cdot \frac{2(a-c)}{3b} = \frac{a + 2c}{3}, \qquad P_N - c = \frac{a-c}{3}.$$

Profit is margin times output, which equals bq_N^2 as the shortcut predicts:

$$\pi_N = \frac{a-c}{3} \cdot \frac{a-c}{3b} = \frac{(a-c)^2}{9b} = \frac{k}{9}.$$

Step 3: the cartel

A monopolist maximises $(a - bQ - c)Q$. The first-order condition is $a - c - 2bQ = 0$ (the second-order condition is $-2b < 0$), so

$$Q^m = \frac{a-c}{2b}, \qquad P^m = a - \frac{a-c}{2} = \frac{a+c}{2}, \qquad P^m - c = \frac{a-c}{2},$$

$$\Pi^m = \frac{a-c}{2} \cdot \frac{a-c}{2b} = \frac{(a-c)^2}{4b} = \frac{k}{4}.$$

Each firm produces half and earns half:

$$q_C = \frac{a-c}{4b}, \qquad \pi_C = \frac{a-c}{2} \cdot \frac{a-c}{4b} = \frac{(a-c)^2}{8b} = \frac{k}{8}.$$

Check: $bq_C^2 = k/16 \neq k/8$. The shortcut fails, as it should, because q_C is not a best response to q_C. Here the margin $(a-c)/2$ exceeds $bq_C = (a-c)/4$, which is exactly why each firm wants to expand.

Step 4: the best deviation, in detail

Suppose the rival sticks to $q_C = \dfrac{a-c}{4b}$. Firm i's best one-period deviation is its best response to that output. Substitute $q_j = q_C$ into BR:

$$q_D = BR(q_C) = \frac{a - c - b \cdot \dfrac{a-c}{4b}}{2b} = \frac{(a-c) - \dfrac{a-c}{4}}{2b} = \frac{\frac{3}{4}(a-c)}{2b} = \frac{3(a-c)}{8b}.$$

- *Size of the deviation.* $q_D = \frac{3}{8}\frac{a-c}{b}$ against $q_C = \frac{2}{8}\frac{a-c}{b}$: the cheater raises its output by 50%. Since $\frac{3}{8} > \frac{1}{3}$, it also produces more than its Cournot output $q_N = \frac{1}{3}\frac{a-c}{b}$. The best response slopes down, so a restrained rival leaves a larger residual demand $a - bq_C - bq_i$ to exploit.
- *Industry output and price.*

$$Q_D = q_D + q_C = \frac{3(a-c)}{8b} + \frac{2(a-c)}{8b} = \frac{5(a-c)}{8b},$$

$$P_D = a - b \cdot \frac{5(a-c)}{8b} = \frac{8a - 5a + 5c}{8} = \frac{3a + 5c}{8}.$$

- *Margin.*

$$P_D - c = \frac{3a + 5c - 8c}{8} = \frac{3(a-c)}{8}.$$

  This equals $b\,q_D = b \cdot \frac{3(a-c)}{8b}$, as the shortcut requires at a best response.

- *Deviator's profit.*

$$\pi_D = (P_D - c)\,q_D = \frac{3(a-c)}{8} \cdot \frac{3(a-c)}{8b} = \frac{9(a-c)^2}{64\,b} = \frac{9k}{64} \qquad (= b\,q_D^2).$$

- *The cheated rival's profit.* The rival still sells q_C, at the deviation margin:

$$\pi_{\text{rival}} = (P_D - c)\,q_C = \frac{3(a-c)}{8} \cdot \frac{a-c}{4b} = \frac{3(a-c)^2}{32\,b} = \frac{6k}{64}.$$

- *Who gains and who loses.* In units of $k/64$: cartel 8 and 8 (joint 16, the monopoly profit $k/4$); deviation 9 and 6 (joint 15). The deviator gains 1, the rival loses 2, and joint profit falls by 1. Cheating is privately profitable but destroys surplus, the prisoner's-dilemma structure of item 2 in general form.

With $a = 100$, $b = 1$, $c = 10$: $q_D = 270/8 = 33.75$, $P_D = 350/8 = 43.75$, margin 33.75, $\pi_D = 1139.0625$ and $\pi_{\text{rival}} = 759.375$, as in item 2 and Figure 1.

Step 5: the three profits on a common denominator

The least common multiple of 9, 8 and 64 is 576:

$$\pi_N = \frac{k}{9} = \frac{64\,k}{576}, \qquad \pi_C = \frac{k}{8} = \frac{72\,k}{576}, \qquad \pi_D = \frac{9k}{64} = \frac{81\,k}{576}.$$

So $\pi_D > \pi_C > \pi_N$ for every admissible (a, b, c). The one-shot gain from cheating is $\pi_D - \pi_C = 9k/576 = k/64$, and the per-period loss from punishment is $\pi_C - \pi_N = 8k/576 = k/72$.

Step 6: the grim-trigger condition, solved directly

The strategy is the one in item 3, with q_C and q_N in place of 22.5 and 30. In the collusion state, the one-stage deviation principle requires (normalised payoffs)

$$\pi_C \;\geq\; (1-\delta)\,\pi_D + \delta\,\pi_N \quad\Longleftrightarrow\quad \frac{72\,k}{576} \;\geq\; (1-\delta)\frac{81\,k}{576} + \delta\,\frac{64\,k}{576}.$$

Multiply both sides by $576/k$. This is allowed, and keeps the direction of the inequality, because $k > 0$:

$$72 \;\geq\; 81(1-\delta) + 64\,\delta = 81 - 17\,\delta \quad\Longleftrightarrow\quad 17\,\delta \geq 9 \quad\Longleftrightarrow\quad \delta \;\geq\; \frac{9}{17}.$$

The same number comes out of the formula of item 3:

$$\delta^* = \frac{\pi_D - \pi_C}{\pi_D - \pi_N} = \frac{(81-72)\,k/576}{(81-64)\,k/576} = \frac{9}{17}.$$

In words: cheating gains 9 units once and costs 8 units in every later period, so the threshold solves $(1-\delta)\cdot 9 = \delta \cdot 8$.

Off the path. After a deviation, both firms produce q_N forever. Against a rival producing q_N, a one-shot deviation x earns at most $\pi_i(BR(q_N), q_N) = \pi_N$, because $BR(q_N) = q_N$. So no deviation is profitable for any δ, whatever (a, b, c).

Step 7: why the parameters disappear

Every profit in the problem has the form (pure number)$\times k$: $\frac{1}{9}k$, $\frac{1}{8}k$, $\frac{9}{64}k$. The intercept a and the cost c enter only through the gap $a - c$, and b only rescales units of quantity. The threshold is a ratio of profit differences, so k cancels. Two very different markets give the same δ^*:

Market	k	q_N	π_N	q_C	π_C	q_D	π_D
$a = 100,\ b = 1,\ c = 10$	8100	30	900	22.5	1012.5	33.75	1139.0625
$a = 50,\ b = 2,\ c = 5$	1012.5	7.5	112.5	5.625	126.5625	8.4375	142.3828125

In the second market,

$$\delta^* = \frac{142.3828125 - 126.5625}{142.3828125 - 112.5} = \frac{15.8203125}{29.8828125} = \frac{9}{17}.$$

What does matter is the *structure*: linear demand, constant marginal cost, two symmetric firms and Cournot punishment. Item 5 changes it.

Answer. With $k = (a-c)^2/b$: $\pi_N = k/9$, $\pi_C = k/8$, and the best deviation $q_D = 3(a-c)/8b$ at price $(3a+5c)/8$ yields $\pi_D = 9k/64$, leaving the rival $6k/64$. On the common denominator 576, the condition is $72 \geq 81 - 17\delta$, so $\delta^* = 9/17$ for every $b > 0$ and $c < a$.

5 Item 5: what makes tacit collusion easier

Item 3 says collusion is sustainable iff

$$\underbrace{(1-\delta)}_{\text{weight on today}}\ \underbrace{(\pi_D - \pi_C)}_{\text{gain from cheating}} \;\leq\; \underbrace{\delta}_{\text{weight on tomorrow}}\ \underbrace{(\pi_C - \pi_N)}_{\text{loss per period}}.$$

Anything that raises δ, lowers the gain from cheating, or raises the loss from punishment helps.

1. Few firms. With n symmetric firms, the Cournot output is $q_N = (a-c)/((n+1)b)$ and $\pi_N = k/(n+1)^2$. The cartel shares the monopoly profit, so $\pi_C = k/(4n)$. The deviator best-responds to $n-1$ rivals each producing $(a-c)/(2nb)$, so $q_D = (n+1)(a-c)/(4nb)$ and $\pi_D = k(n+1)^2/(16n^2)$. Then the gain is $\pi_D - \pi_C = k(n-1)^2/(16n^2)$, the loss is $\pi_C - \pi_N = k(n-1)^2/(4n(n+1)^2)$, and

$$\frac{\text{gain}}{\text{loss}} = \frac{(n+1)^2}{4n}, \qquad \delta^*(n) = \frac{\pi_D - \pi_C}{\pi_D - \pi_N} = \frac{(n+1)^2}{n^2+6n+1}.$$

n	2	3	4	5	6	7	8	9	10
δ^*	$\frac{9}{17}$	$\frac{4}{7}$	$\frac{25}{41}$	$\frac{9}{14}$	$\frac{49}{73}$	$\frac{16}{23}$	$\frac{81}{113}$	$\frac{25}{34}$	$\frac{121}{161}$
$\approx$	0.529	0.571	0.610	0.643	0.671	0.696	0.717	0.735	0.752

$\delta^*(n)$ is increasing in n and tends to 1. In the inequality: the ratio gain/loss grows with n, because the punishment loss $\pi_C - \pi_N$ shrinks fast (Cournot with many firms is already close to competitive) while the temptation to undercut a cartel that restrains output stays large. Concentration and entry barriers facilitate collusion; entry or a competitive fringe breaks it.

2. Frequent interaction. Write $\delta = e^{-r\Delta t}$, where r is the interest rate and Δt the time between decisions. The duopoly condition $\delta \geq 9/17$ is $r\Delta t \leq \ln(17/9) \approx 0.636$. Shorter periods (frequent, repeated sales), a low interest rate and a high probability that the market continues (slide 22: δ also reflects the chance the relationship ends) all raise δ. Infrequent, lumpy sales (one-off auctions, long gaps between orders) are hard to collude in.

3. Observability and speed of detection. The strategy needs each firm to see the deviation and to retaliate. If detection takes several periods, the deviator enjoys π_D for all of them, so the effective gain from cheating in item 3 rises. Posted prices and public sales data shorten the lag; secret discounts and hidden quantities lengthen it. With imperfect monitoring (Green–Porter), a low price may reflect a demand shock rather than cheating, so firms must punish on bad signals and price wars occur on the equilibrium path; this lowers the value of collusion below π_C forever.

4. Demand growth or decline. Growing demand raises the profit stakes of later periods, which acts like a higher δ: the future loss from punishment is larger relative to today's gain, so collusion is easier. Declining demand, or a market near its end, does the opposite; a known last period makes cooperation unravel (Lecture 5, finitely repeated games). Temporary demand booms raise today's gain $\pi_D - \pi_C$ relative to the future, so collusion is most fragile in booms.

5. Symmetry. Item 3 assumed identical firms. When firms differ in costs, capacity or patience, each has its own threshold and the cartel needs all of them satisfied; the firm with the largest deviation gain or the lowest δ binds. Similar costs, products and shares make a common cartel output and punishment natural.

6. Multimarket contact. If the same firms meet in several markets, a deviation in one can be punished in all, so the loss in the right-hand side of the inequality is larger. Slack in one market's constraint also supports a market where the constraint binds.

7. Capacity. Spare capacity in the hands of rivals makes punishment credible and costly (a larger loss), but spare capacity in the hands of a potential cheater makes cheating cheap and large (a larger gain). Collusion is easiest when capacity is similar and roughly matched to the cartel output.

Remark: softer punishment. Nash reversion is not the harshest credible punishment. Abreu's carrot-and-stick strategies (a short, severe punishment phase, then a return to collusion) can drive the punishment payoff below π_N for a while, which lowers the threshold below 9/17 for the same cartel output. I give no number for this; the script does not verify one.

References

- Lecture 5 slides (`Aula/Lecture05.pdf`): slide 21 (normalised payoffs), slide 22 (interpreting discounting), slide 25 (one-stage deviation principle), slides 28–29 (grim trigger), slide 34 (this homework).
- Tadelis, S. (2013), *Game Theory: An Introduction*, Princeton University Press, Section 10.4 "Application: Tacit Collusion", p. 201 (the same Cournot example).
- Cabral, L. (2017), *Introduction to Industrial Organization*, 2nd ed., MIT Press, Chapter 9 "Collusion and Price Wars", Section 9.3 "Factors that facilitate collusion", p. 227.
- Gibbons, R. (1992), *Game Theory for Applied Economists*, Princeton University Press, Section 2.3.C "Collusion between Cournot Duopolists", p. 102.
- Green, E. and Porter, R. (1984), "Noncooperative collusion under imperfect price information", *Econometrica* 52(1); Abreu, D. (1986), "Extremal equilibria of oligopolistic supergames", *Journal of Economic Theory* 39(1): cited from memory, without pages.